

Introduction to Environmental Science EES102

Lecture by: Baerbel Sinha, IISERM

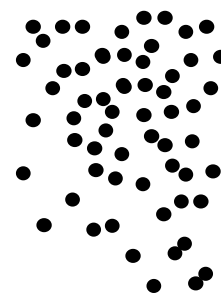
ATMOSPHERIC AEROSOL PARTICLES

PRIMARY AND SECONDARY AEROSOL PARTICLES

- Definition: **Primary aerosol particles** are emitted directly as particles
- Definition: **Secondary aerosol particles** are formed in the atmosphere from gaseous emissions through gas to particle conversion



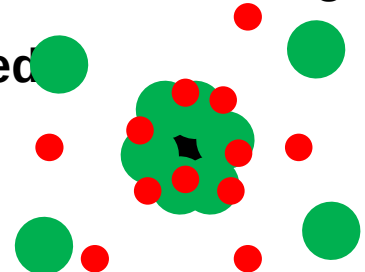
Condensable vapour



freshly nucleated particle
~ nm



Condensation and coagulation



ATMOSPHERIC AEROSOL PARTICLES

NATURAL VERSUS ANTHROPOGENIC SOURCE

- Definition: Aerosol particles from natural sources are emitted without human intervention
- Definition: Anthropogenic aerosol particles are man-made or emitted because human intervention



Sources	Approx. global emission flux (Tg a ⁻¹)
<i>Primary aerosols</i> Desert and semi-arid areas (mineral dust) Marine sea spray Volcanic dust Biogenic (pollens, debris) <i>Secondary aerosols</i> Sulphates from biogenic gases Sulphates from volcanic SO₂ Nitrates from nitrogen oxides Organics from volatile organic compounds (COV) TOTAL	1500 1300 33 50 90 12 55 22 ~3060
<i>Primary aerosols</i> Industrial dust Soot <i>Secondary aerosols</i> Organics from oxidation of volatile organic compounds Sulphates produced by oxidation of anthropogenic SO₂ Sulphates produced from biomass combustion	100 10 10 190 90 50

If the natural burden is so high ($\sim 3060 \text{ Tg an}^{-1}$) and the anthropogenic contribution only slightly more than 13% of that (390 Tg an^{-1}), why do we attempt to reduce the aerosol burden? We can't do anything about the natural background, right?

➤ **The chemical composition is different!**
Inhalation of sea salt aerosol is a treatment for many lung conditions (Chronic bronchitis, Asthma, ...) => Aerosol emitted by traffic can be their cause.

➤ **The short lifetime of aerosol means the aerosol burden and its effects are highly local. Anthropogenic emissions are where we are!**

Silicosis: Occupational lung disease

Silicosis is an often fatal lung disease caused by breathing dust containing crystalline silica particles, a basic component of sand and granite. There is no cure for silicosis, and treatment options are limited. However, the condition can be prevented if measures are taken to reduce exposure.

Symptoms

Continued exposure:

- Shortness of breath
- Fever
- Bluish skin at the ear lobes or lips

As the disease progresses:

- Fatigue
- Extreme shortness of breath
- Loss of appetite
- Chest pain
- Respiratory failure

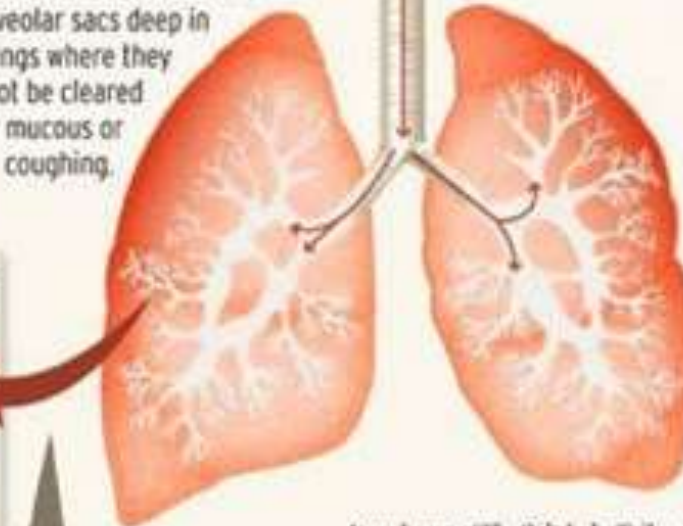
At-risk occupations

- Construction
- Mining
- Sandblasting
- Masonry
- Demolition
- Manufacturing of glass and metal products
- Plumbing
- Painting

CRYSTALLINE SILICA DUST

Inhaling the dust can cause scar tissue to form in the lungs that reduces the lungs' ability to extract oxygen from the air.

Silica dust particles can embed themselves in the alveolar sacs deep in the lungs where they cannot be cleared by mucus or coughing.



AMY LEWIS/The Salt Lake Tribune

Safety instruction!

Do wear masks while exposed to dust

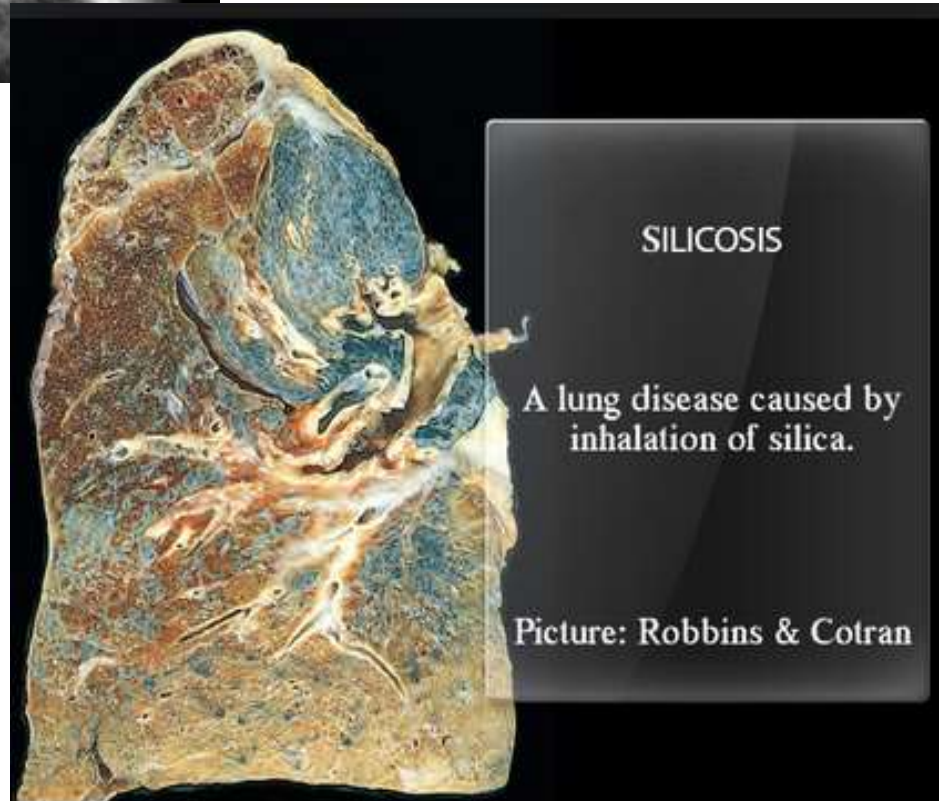


**X-ray of lung
with silicosis**

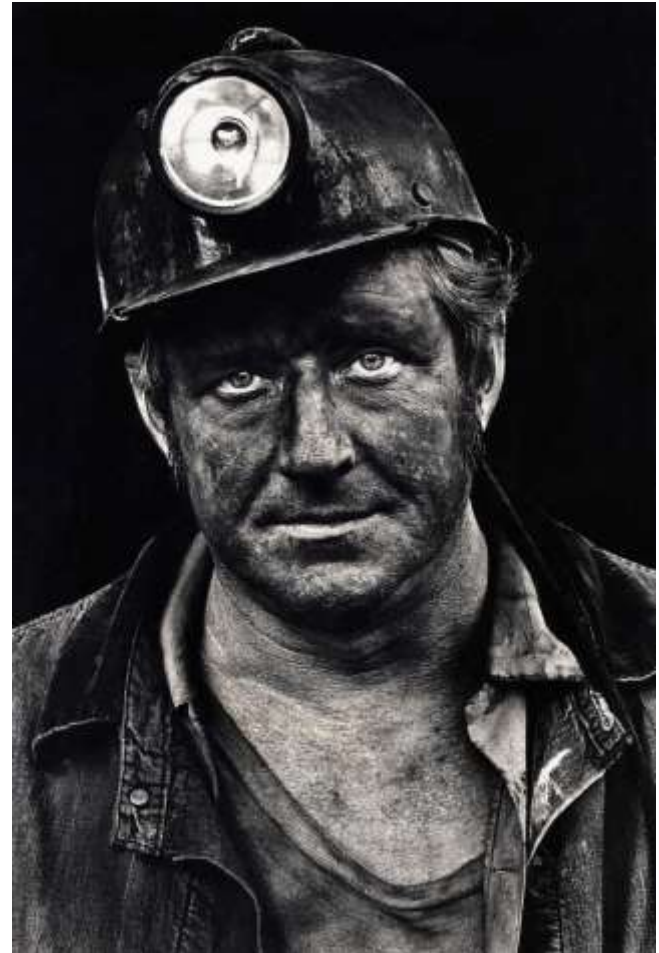
**Coal
miners
lung after
autopsy**



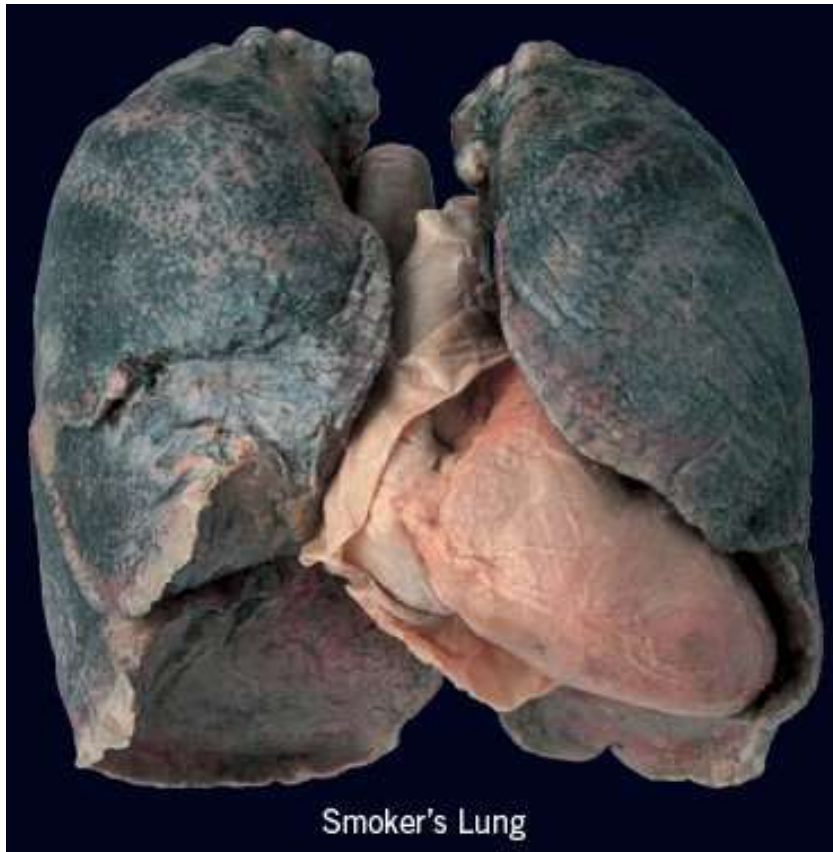
**Silicosis
lung
after
autopsy**



Miners lung



SMOKERS LUNG



While BC can decompose into primary spheres and can enter the blood stream (and does), there are limits to how much BC the body processes that way.

A chain smoker (and females cooking with poor ventilation in particular when cooking using biomass burning stoves) will develop a smokers lung over time

The main difference between the lung of a coal miner and a chain smoker is that the smokers lung is blackest in the alveoli, while the miner has more dust in the bronchiolar airways

➤ **The size distribution is different!**

➤ Sea salt and dust have large aerodynamic diameters mostly $>2.5\ \mu\text{m}$

➤ When we inhale them, most of the particulate is lost upwards of the larynx and removed from the body via the mucus escalator

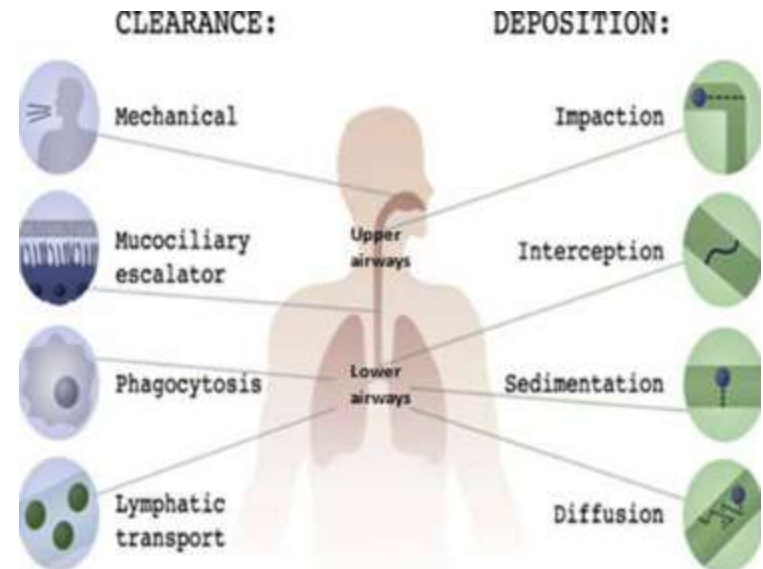
➤ Anthropogenic aerosols are **very small**

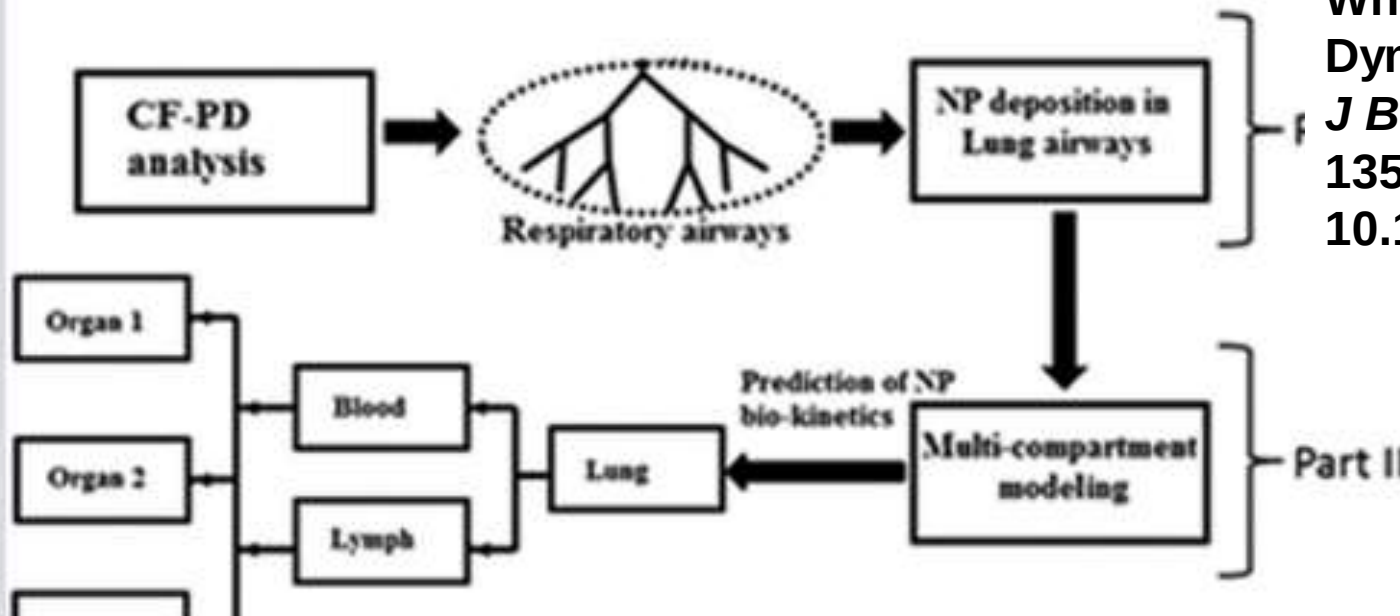
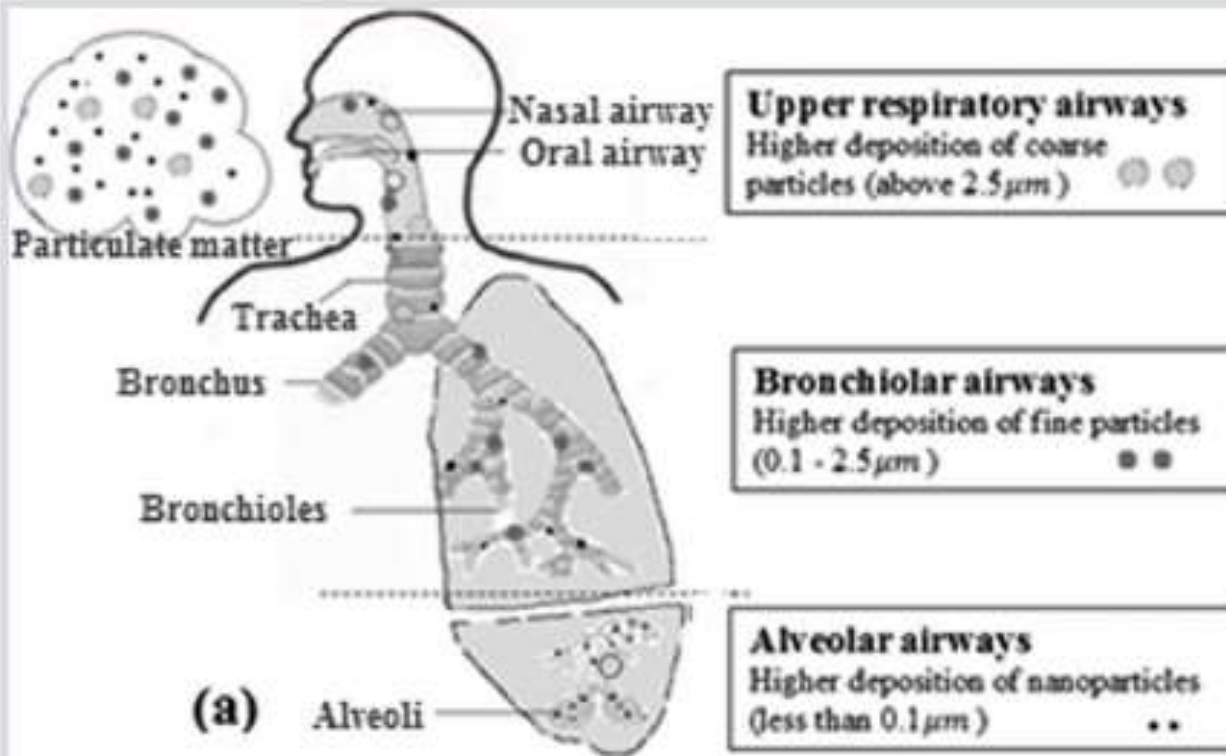
➤ Typically they have two modes one at $\sim 40\text{-}80\ \text{nm}$ and one at $\sim 150\text{-}300\ \text{nm}$

➤ They travel into the alveoli where cleaning them up can take months to years.

➤ **They decompose into primary spheres (soot) that enter the blood stream which promote cardiovascular failures or are eaten by macrophages & translocated.**

➤ **If all fails (small dust) they stick around forever causing a condition called silicosis**



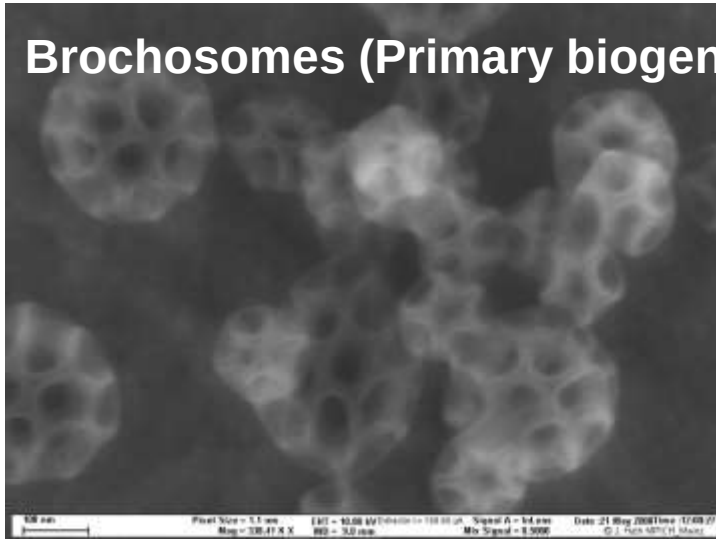


Arun V. Kolanjiyil
and Clement
Kleinstreuer,
Nanoparticle Mass
Transfer From Lung
Airways to Systemic
Regions—Part I:
Whole-Lung Aerosol
Dynamics

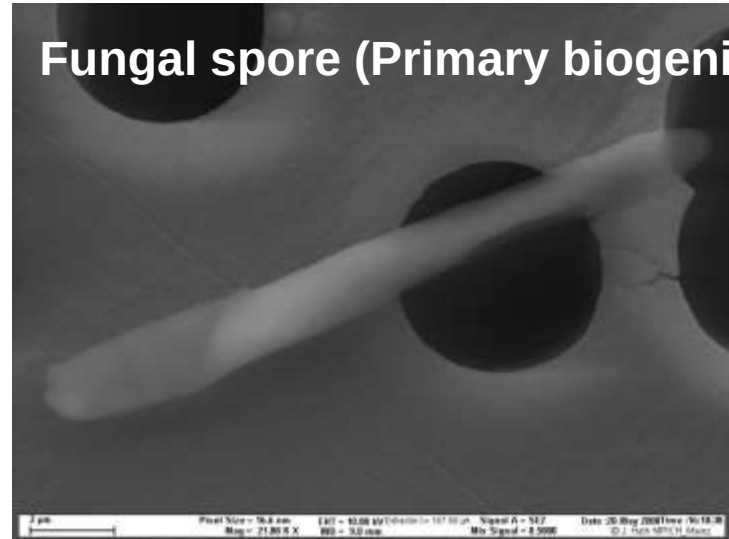
J Biomech Eng
135(12), 121003, doi:
10.1115/1.4025332

SIZE IS IMPORTANT BUT HOW TO MEASURE IT

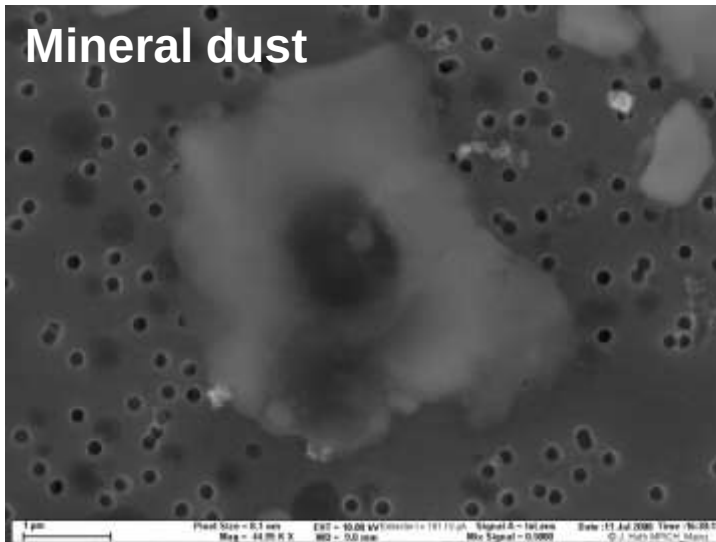
Brochosomes (Primary biogen



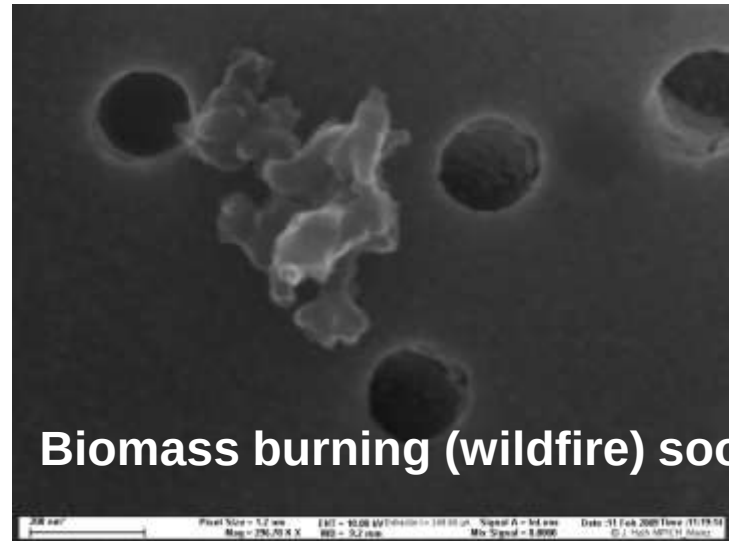
Fungal spore (Primary biogeni

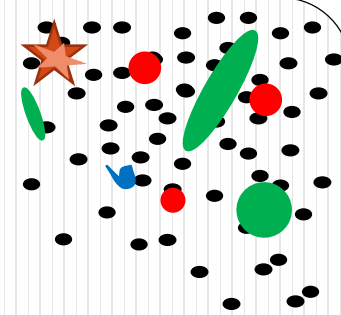


Mineral dust



Biomass burning (wildfire) soot



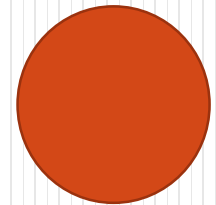


Aerodynamic equivalent diameter

Diameter of a unit density sphere having the same gravitational settling velocity as the particle in question

Unit density = 1 g/cm^3

In other words the aerodynamic equivalent diameter of a $4 \text{ }\mu\text{m}$ spherical bacterial cell with a density of 1 g/cm^3 is $4 \text{ }\mu\text{m}$



INDOOR AIR POLLUTION

Gasses: CO, Formaldehyde

Particles: Asbestos, microplastics

Solid fuel usage (NO, CO, VOCs, PM)

Gas stove usage (NO, CO, VOCs, PM)

Radioactive Radon

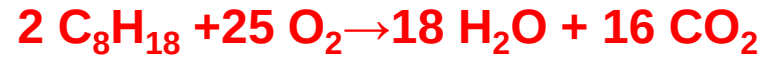
Ventilation reduces indoor air pollution

Sealing the basement reduces the transfer of Radon from the soil to the house

Choose your interior & insulation – no asbestos as little synthetic material as possible

ENGINES/POWER PLANTS AND THEIR EMISSIONS

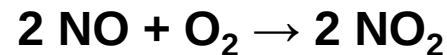
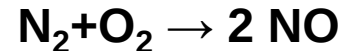
Fuel lean combustion lowers CO because of



Fuel rich combustion maximizes CO & power output

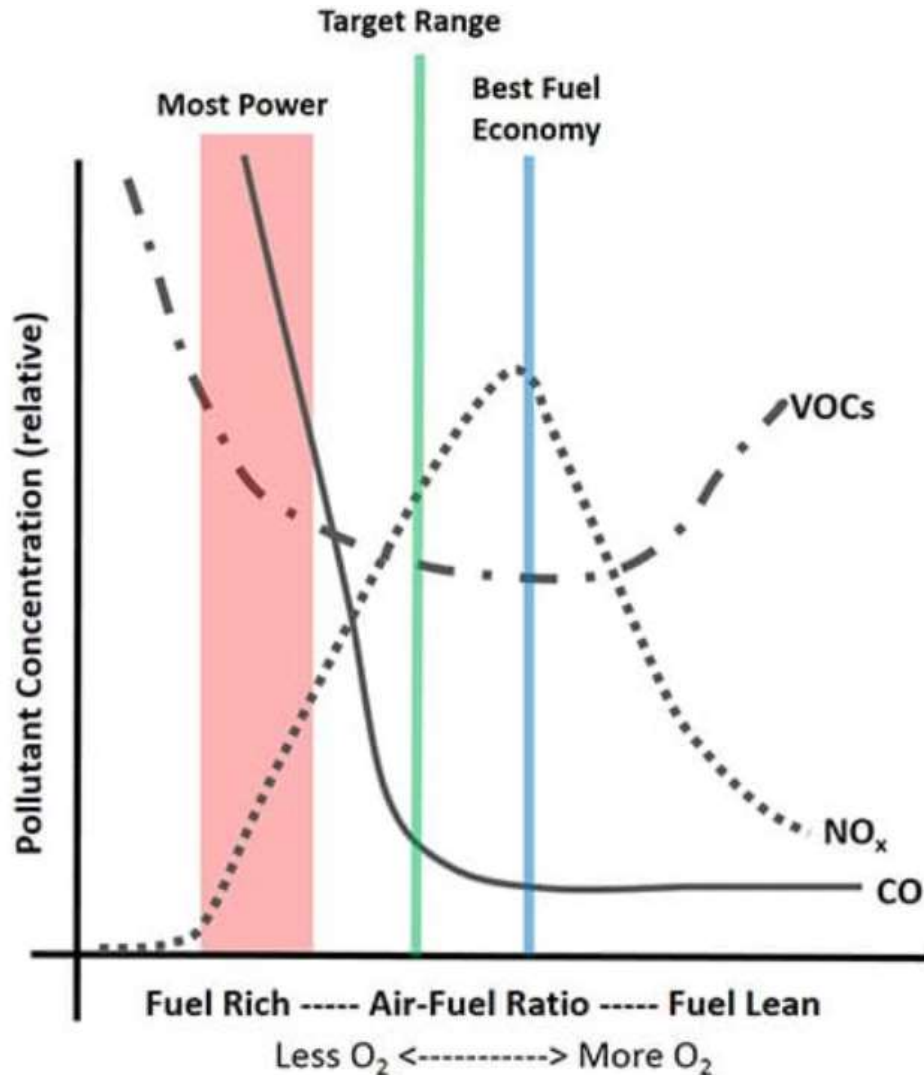


The best fuel economy (miles/L) coincides with the highest combustion temperature maximum NO_x emissions



Reduced VOCs (unburnt fuel) are most abundant during fuel rich combustion

Fuel lean combustion maximizes OVOC emissions



ENGINES/POWER PLANTS AND THEIR EMISSIONS

Temperature and therefore NO emissions depend on:

1. How many molecules are burnt (how much fuel is ignited by the spark)
2. How much energy is generated per molecule burnt (CO_2 more than 2x CO)
3. How much (usually cool) air is fed into the flame

High temperature is needed because it consumes energy to break the N_2 triple bond and make N_2 react with O_2 . Reaction of NO with O_2 also consumes energy

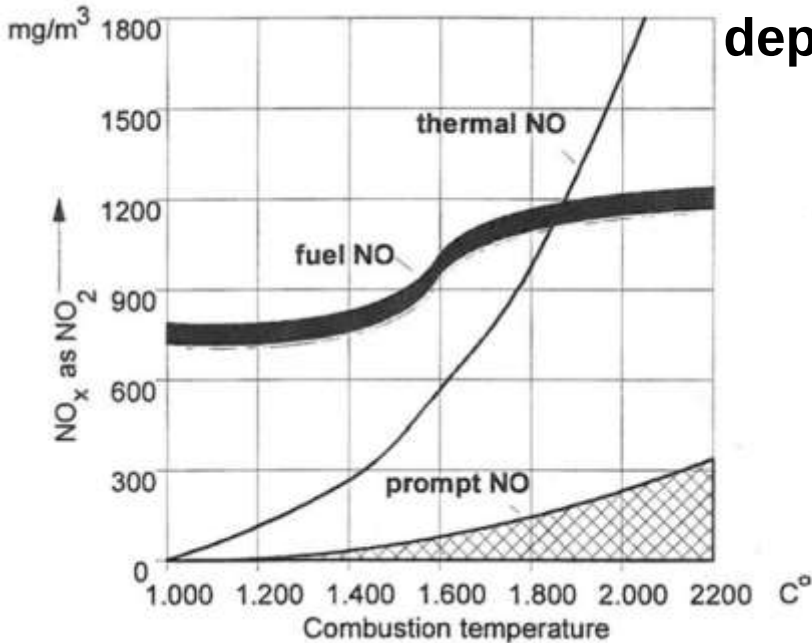
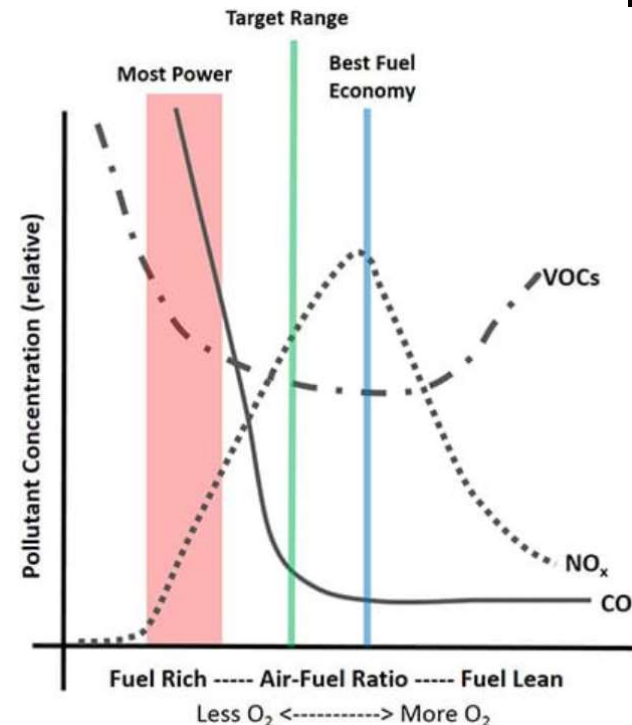
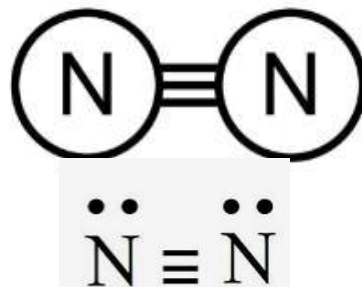
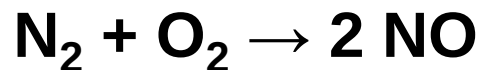


TABLE 7.1 STANDARD (25°C) ENTHALPIES AND FREE ENERGIES OF FORMATION FOR SOME ATMOSPHERIC COMPOUNDS (kJ)

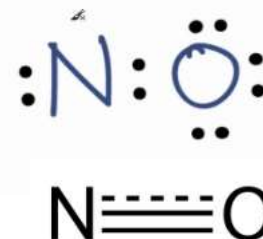
	ΔH_0	ΔG_0
O_3	142.2	163.4
CO_2	-393.4	-394.3
CO	-110.5	-137.2
NO	90.4	86.7
NO_2	33.8	51.8
$\text{HNO}_3(\text{aq})$	-206.5	-110.5
SO_2	-296.8	-300.3
SO_3	-395.1	-370.3
$\text{H}_2\text{SO}_4(\text{aq})$	-907.3	-741.8
$\text{H}_2\text{O}(\text{g})$	-241.8	-228.6
$\text{H}_2\text{O}(\text{l})$	-285.7	-237.2



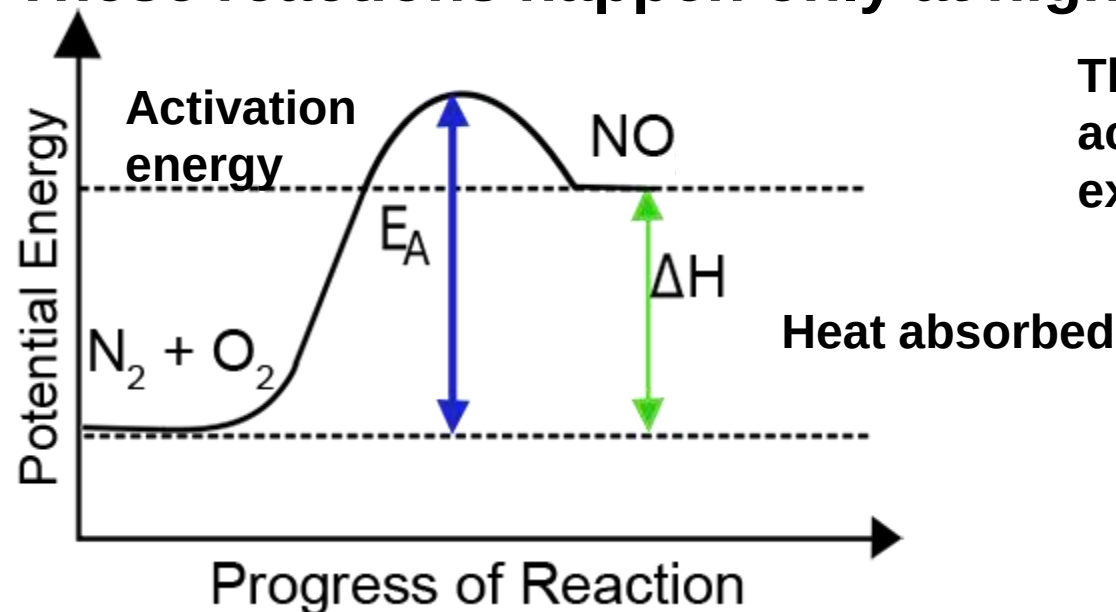
A CLOSER LOOK AT NITROGEN OXIDES IN THE ATMOSPHERE



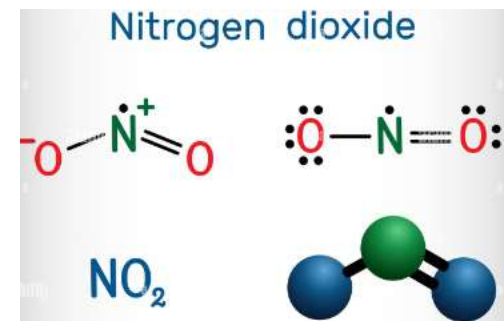
Nitrogen Monoxide
(Nitric Oxide)



These reactions happen only at high temperature

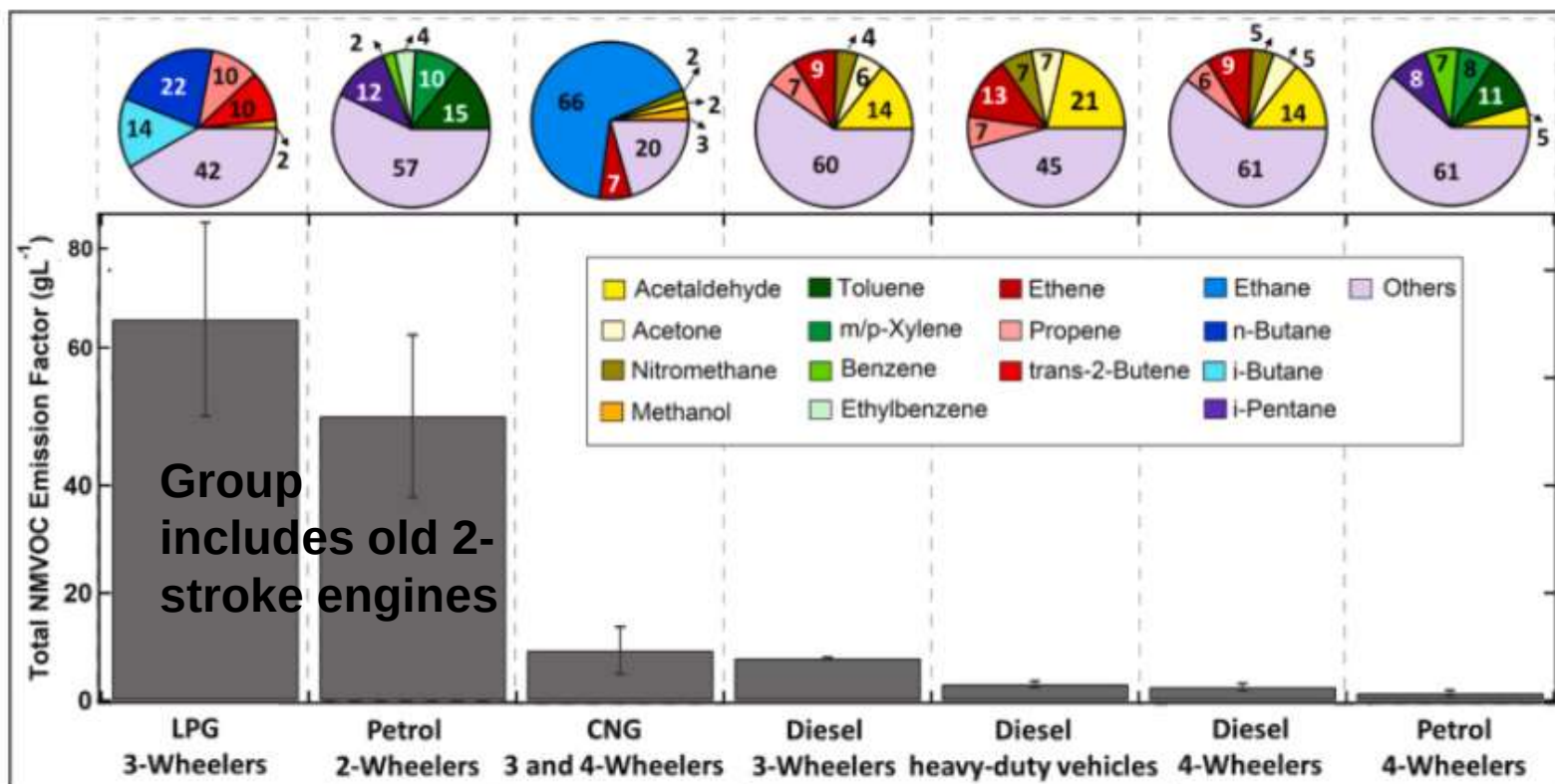
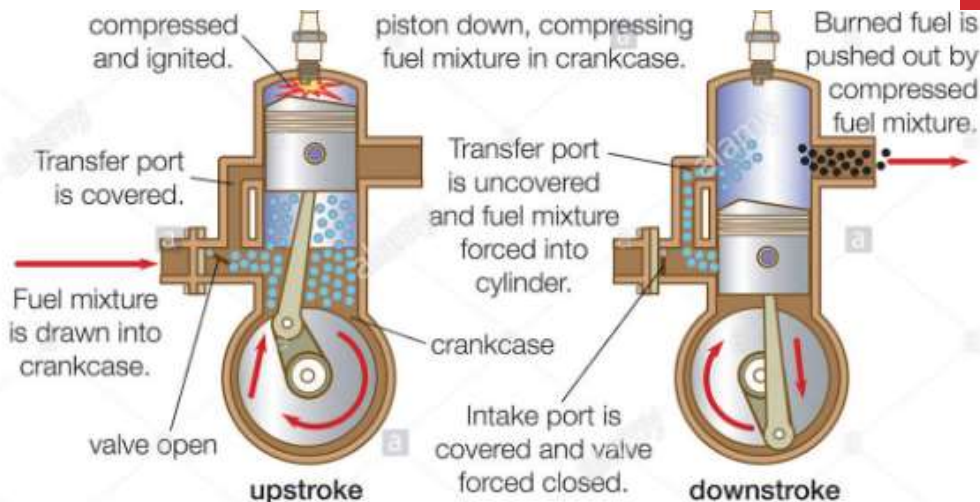


The second reaction requires activation energy but is exothermic



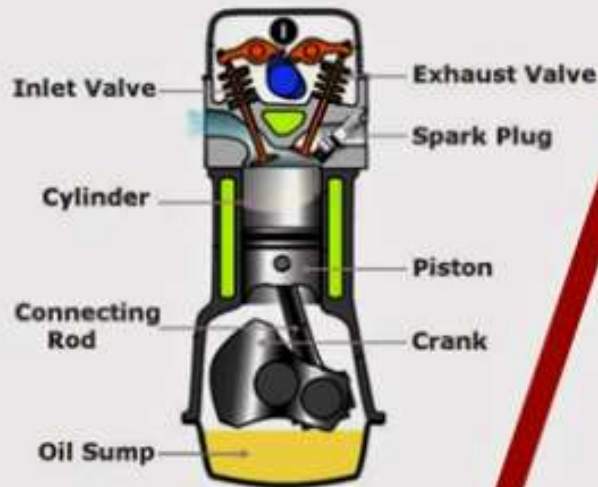
ENGINES AND THEIR VOC EMISSIONS

HAKKIM ET AL. ATMOSPHERIC ENVIRONMENT X, 2022

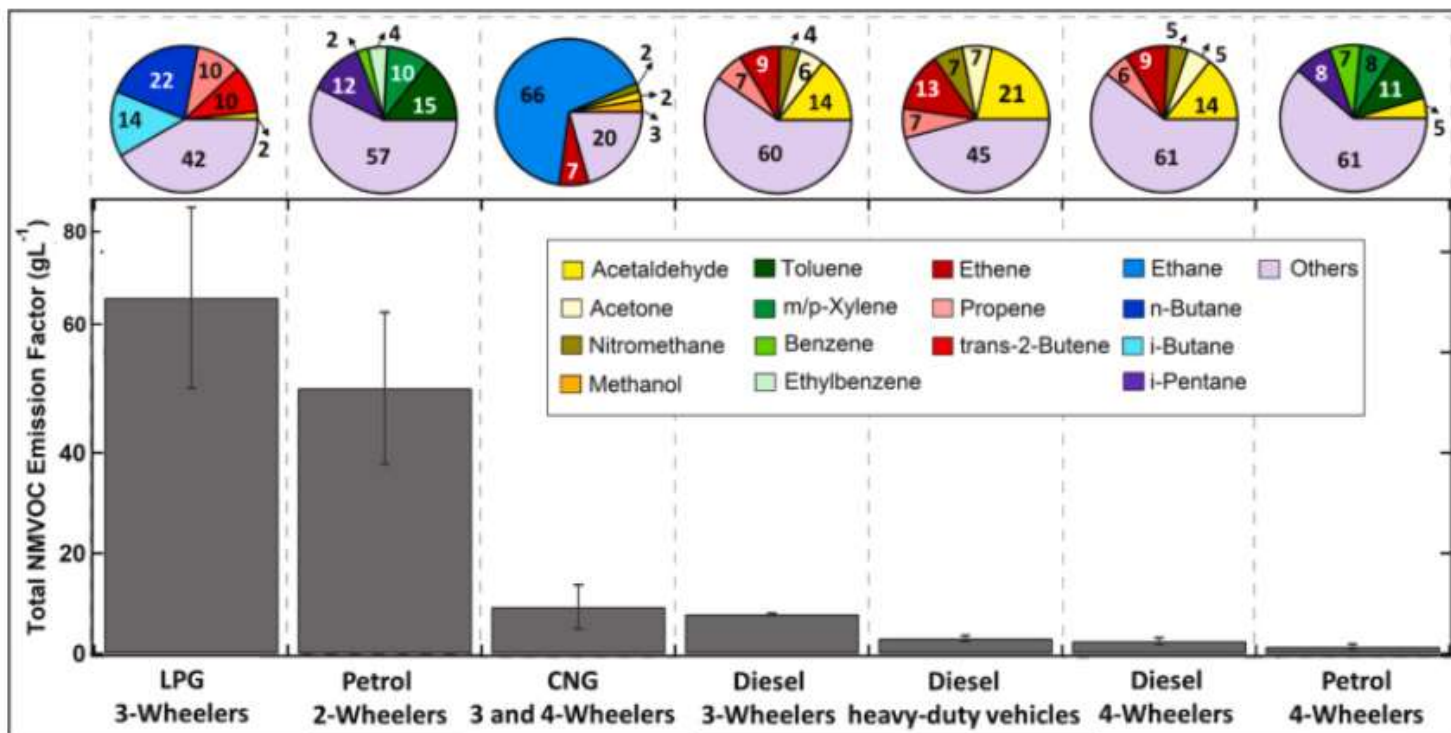
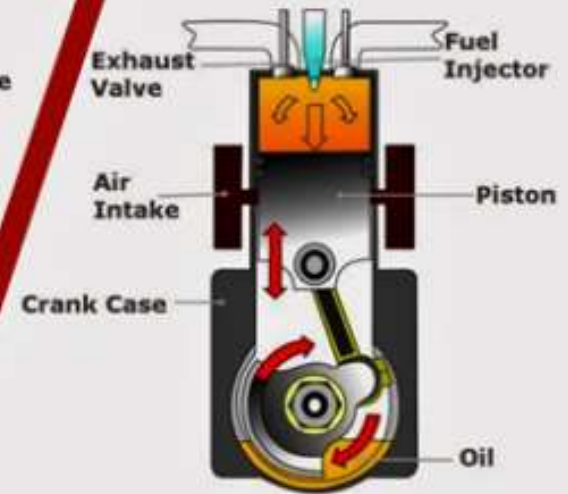


ENGINES AND THEIR VOC EMISSIONS

PETROL ENGINE

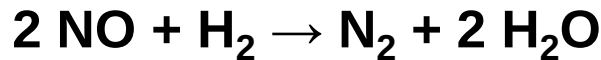
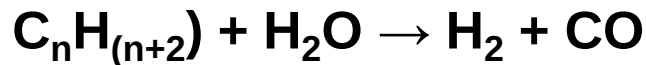


DIESEL ENGINE

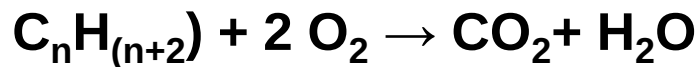
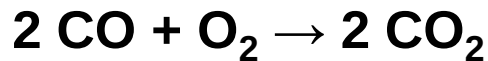


CATALYTIC CONVERTERS IN CARS

Chamber 1 Ceramic honeycomb coated with platinum Pt, Paladium PD and rhodium Rh



Air injection + Pt catalyst



Cars with catalytic converters need to be adjusted to a very specific air/fuel ratio 14.8/1 to optimize the working of the catalyst. Too much air means higher NOx emissions. Too much fuel means high CO and VOC emissions.

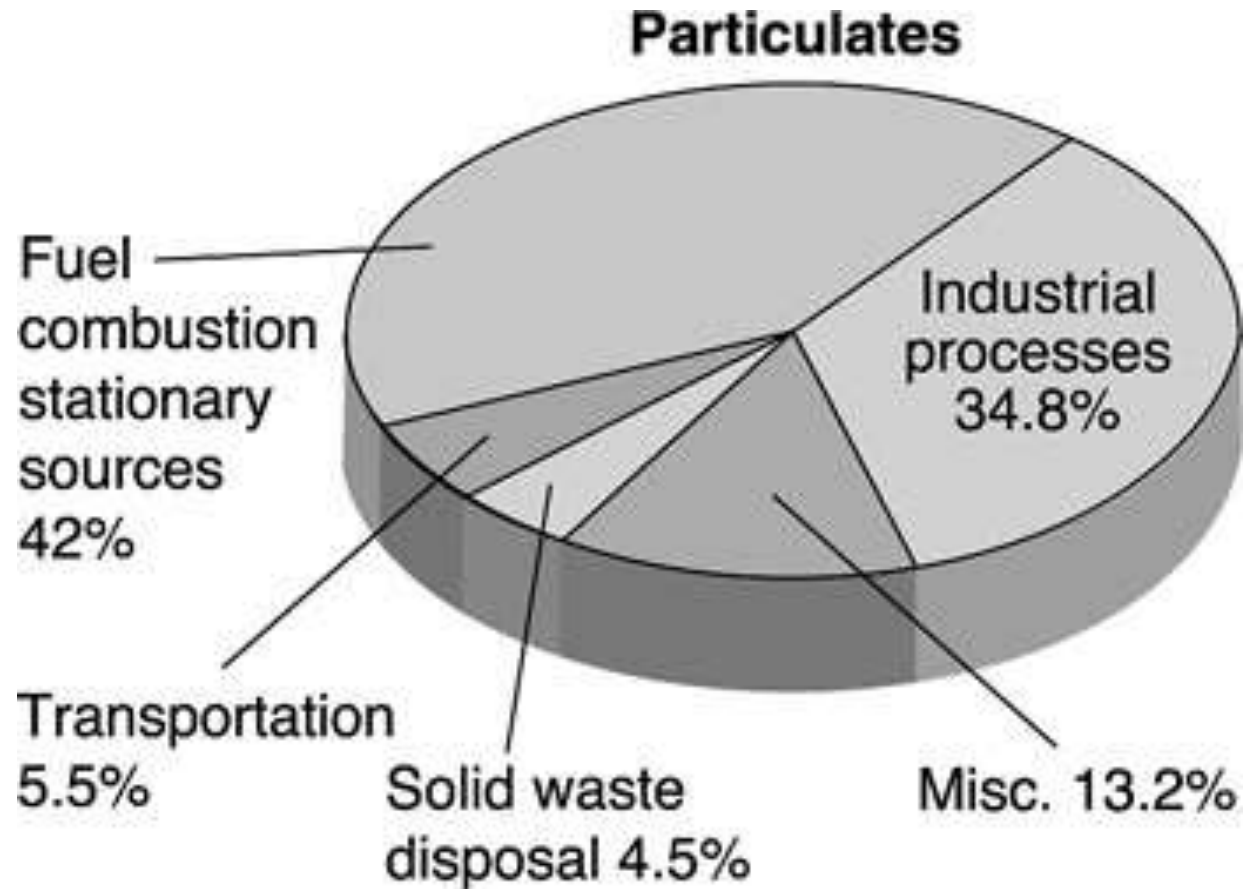
**They only work if the temperature is 350°C to 600°C.
Short trips => unmitigated emissions**

NOX CONTROL IN POWER PLANTS

Power plants/boilers often control for NO_x emissions by recycling hot flue gas back into the fire to make the burning zone burning more fuel, and pyrolysis gas rich but hot . Pyrolysis gas has H₂

Surplus air is then added further up in the flame to have a low temperature fuel rich combustion to minimize CO emissions

Figure 2.9 US nationwide emissions of particulates



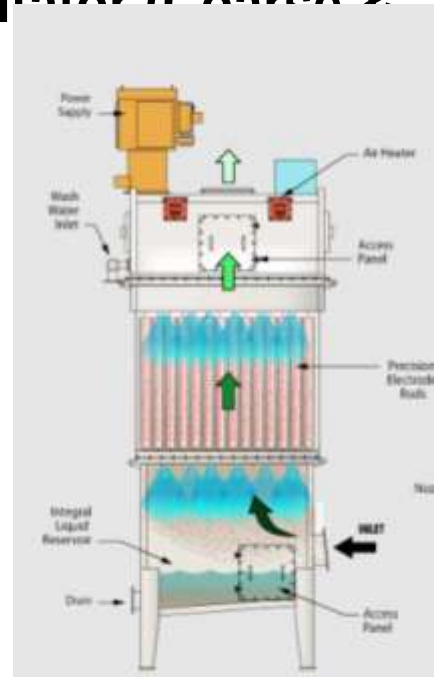
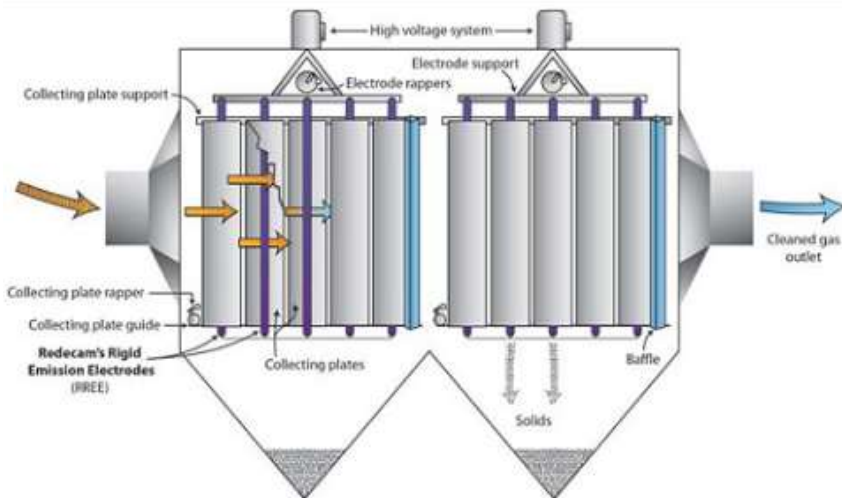
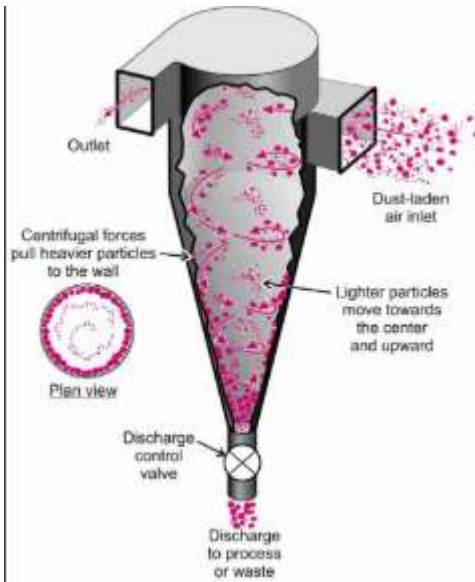
PM CONTROL IN POWER PLANTS

Particles are removed using
Cyclones (coarse mode only)

Filters (coarse mode only)

Electrostatic precipitator (coarse & fine)

Wet electrostatic precipitator (Coarse & fine particles & SO₂)



PM CONTROL IN POWER PLANTS

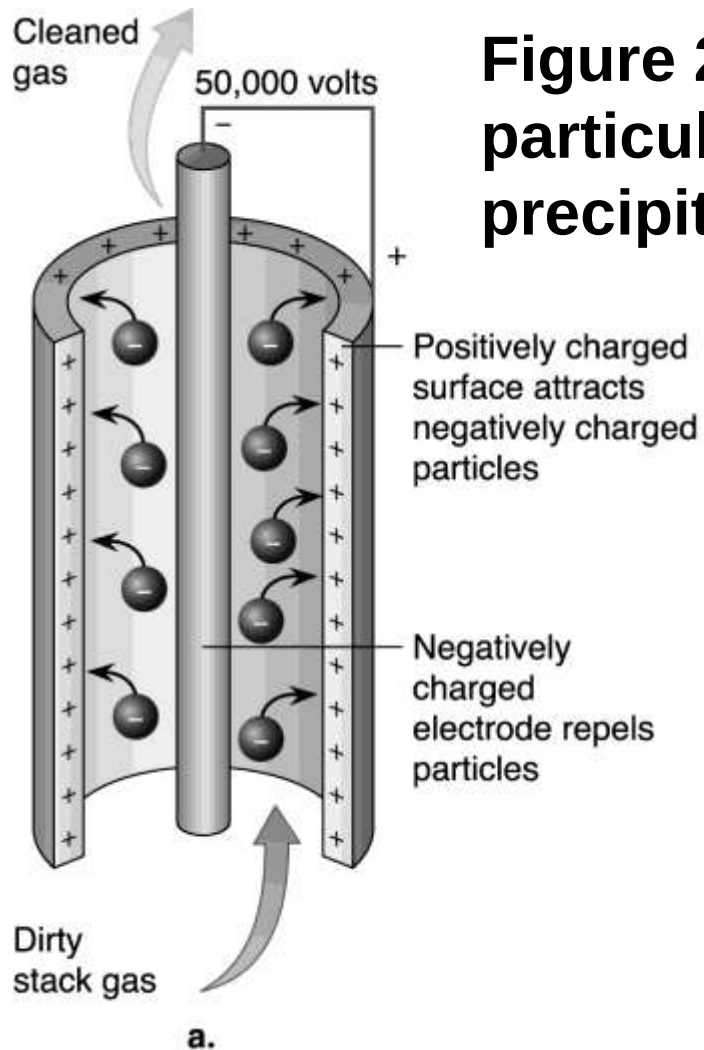


Figure 2.10 Two methods for controlling particulate emissions. (a) electrostatic precipitation, (b) bag filtration

